

Linköping Studies in Science and Technology. Dissertations
No. 1256

Control of EGR and VGT for Emission Control and Pumping Work Minimization in Diesel Engines

Johan Wahlström

Division of Vehicular Systems
Department of Electrical Engineering
Linköping University, SE-581 83 Linköping, Sweden

Linköping 2009

**Control of EGR and VGT for
Emission Control and
Pumping Work Minimization in
Diesel Engines**

© 2009 Johan Wahlström

johwa@isy.liu.se
<http://www.vehicular.isy.liu.se>
Department of Electrical Engineering,
Linköping University,
SE-581 83 Linköping,
Sweden.

ISBN 978-91-7393-611-8

ISSN 0345-7524

Printed by LiU-Tryck, Linköping, Sweden 2009

Abstract

Legislators steadily increase the demands on lowered emissions from heavy duty vehicles. To meet these demands it is necessary to integrate technologies like Exhaust Gas Recirculation (EGR) and Variable Geometry Turbochargers (VGT) together with advanced control systems. Control structures are proposed and investigated for coordinated control of EGR valve and VGT position in heavy duty diesel engines. Main control goals are to fulfill the legislated emission levels, to reduce the fuel consumption, and to fulfill safe operation of the turbocharger. These goals are achieved through regulation of normalized oxygen/fuel ratio and intake manifold EGR-fraction. These are chosen as main performance variables since they are strongly coupled to the emissions.

To design successful control structures, a mean value model of a diesel engine is developed and validated. The intended applications of the model are system analysis, simulation, and development of model-based control systems. Dynamic validations show that the proposed model captures the essential system properties, i.e. non-minimum phase behaviors and sign reversals.

A first control structure consisting of PID controllers and min/max-selectors is developed based on a system analysis of the model. A key characteristic behind this structure is that oxygen/fuel ratio is controlled by the EGR-valve and EGR-fraction by the VGT-position, in order to handle a sign reversal in the system from VGT to oxygen/fuel ratio. This structure also minimizes the pumping work by opening the EGR-valve and the VGT as much as possible while achieving the control objectives for oxygen/fuel ratio and EGR-fraction. For efficient calibration an automatic controller tuning method is developed. The controller objectives are captured by a cost function, that is evaluated utilizing a method choosing representative transients. Experiments in an engine test cell show that the controller achieves all the control objectives and that the current production controller has at least 26% higher pumping losses compared to the proposed controller.

In a second control structure, a non-linear compensator is used in an inner loop for handling non-linear effects. This compensator is a non-linear state dependent input transformation. PID controllers and selectors are used in an outer loop similar to the first control structure. Experimental validations of the second control structure show that it handles nonlinear effects, and that it reduces EGR-errors but increases the pumping losses compared to the first control structure.

Substantial experimental evaluations in engine test cells show that both these structures are good controller candidates. In conclusion, validated modeling, system analysis, tuning methodology, experimental evaluation of transient response, and complete ETC-cycles give a firm foundation for deployment of these controllers in the important area of coordinated EGR and VGT control.

Sammanfattning

Lagkrav på emissioner för tunga fordon blir allt hårdare samtidigt som man vill ha låg bränsleförbrukning. För att kunna möta dessa krav införs nya teknologier såsom återcirkulering av avgaser (EGR) och variabel geometri-turbin (VGT) i dieselmotorer. I EGR-systemet finns ett spjäll som gör att man kan påverka EGR-flödet och i VGT:n finns ett ställdon som gör att man kan påverka turbinflödet. De primära mekanismerna som används för att minska emissioner är att kväveoxider kan minskas genom att öka andelen EGR-gaser i insugsröret, och att partiklar kan minskas genom att öka syre/bränsle-förhållandet i cylindrarna. Därför väljes EGR-andel och syre/bränsle-förhållande som prestandavariabler. Dessa prestandavariabler beror på ett komplicerat sätt av positionerna i EGR-spjället och i VGT:n, och det är därför nödvändigt att ha samtidig reglering av EGR och VGT för att uppnå lagkraven på emissioner.

För att designa framgångsrika reglerstrukturer, utvecklas och valideras en matematisk modell av en dieselmotor. Modellen används för systemanalys, simulering och utveckling av modellbaserade reglersystem. Dynamiska valideringar visar att den föreslagna modellen fångar de väsentliga systemegenskaperna, vilka är icke-minfasbeteenden och teckenväxlingar.

En första reglerstruktur som består av PID-regulatorer och min/max-väljare är utvecklad baserat på en systemanalys av modellen. Huvudlooparna i strukturen väljes så att syre/bränsle-förhållandet regleras av EGR-spjället och EGR-andelen regleras av VGT-positionen för att hantera en teckenväxling i systemet från VGT till syre/bränsle-förhållande. Denna struktur minimerar också bränsleförbrukningen genom att minimera pumpförluster, där pumpförluster orsakas av att trycket på avgassidan är större än trycket på insugssidan i en stor del av arbetsområdet. Principen i denna minimering är att öppna EGR-spjället och VGT:n så mycket som möjligt under tiden som reglermålen för syre/bränsle-förhållande och EGR-andel är uppfyllda. För att få en effektiv kalibrering av reglerstrukturen utvecklas en automatisk inställningsmetod av regulatorparametrarna. Reglermålen fångas av en kostnadsfunktion, som utvärderas genom att använda en metod för att välja ut representativa transienter. Experiment i en motortestcell visar att regulatorn klarar av alla reglermål och att den nuvarande regulatorn som finns i produktion har minst 26% högre pumpförluster jämfört med den föreslagna regulatorn.

I en andra reglerstruktur används en olinjär kompensator i en inre loop för att hantera olinjära effekter. Denna kompensator är en olinjär tillståndsberoende transformation av insignaler. PID-regulatorer och väljare används i en yttre loop på liknande sätt som för den första reglerstrukturen. Experiment med den andra reglerstrukturen visar att den hanterar olinjära effekter, och att den minskar EGR-fel men ökar pumpförlusterna jämfört med den första reglerstrukturen.

Omfattande experimentella utvärderingar i motortestceller visar att båda dessa regulatorstrukturer är goda kandidater. Sammanfattningsvis ger modellering, systemanalys, inställningsmetodik, experimentella utvärderingar av transientsvar och fullständiga europeiska transientcykler en stabil grund för användning av dessa regulatorer vid samtidig reglering av EGR och VGT.

Acknowledgments

This work has been performed at the department of Electrical Engineering, division of Vehicular Systems, Linköping University, Sweden. I am grateful to my professor and supervisor Lars Nielsen for letting me join this group, for all the discussions we have had, and for proofreading my work.

I would like to thank my second supervisor Lars Eriksson for many interesting discussions, for giving valuable feedback on the work, and for telling me how to improve my research. Thanks go to Erik Frisk for the discussions regarding my research and the help regarding LaTeX. Carolina Fröberg, Susana Högne, and Karin Bogg are acknowledged for all their administrative help and the staff at Vehicular Systems for creating a nice working atmosphere.

I also thank Magnus Pettersson, Mats Jennische, David Elfvik, David Vestgöte, and Yones Strand at Scania CV AB for the valuable meetings, for showing great interest, and for the measurement supply. Also the Swedish Energy Agency are gratefully acknowledged for their financial support.

A special thank goes to Johan Sjöberg for being a nice friend, for getting me interested in automatic control and vehicular systems during the undergraduate studies, and for giving me a tip of a master's thesis project at Vehicular Systems

Finally, I would like to express my gratitude to my parents, my sister, my brother, and Kristin for always being there and giving me support and encouragement.

Linköping, April 2009

Johan Wahlström

Contents

I	Introduction	1
1	Introduction	3
1.1	List of Publications	5
1.2	Overview and Contributions of the Publications	6
1.2.1	Publication 1 - Modeling	6
1.2.2	Publication 2 - System analysis	8
1.2.3	Publication 3 - EGR-VGT Control for Pumping Work Min- imization	9
1.2.4	Publication 4 - Controller Tuning	10
1.2.5	Publication 5 - Non-linear compensator	11
1.2.6	Publication 6 - Non-linear control	11
	Bibliography	13
II	Publications	17
1	Modeling of a Diesel Engine with VGT and EGR capturing Sign Reversal and Non-minimum Phase Behaviors	19
1	Introduction	21
1.1	Outline and model extensions	21
1.2	Selection of number of states	22
1.3	Model structure	22
1.4	Measurements	23

1.5	Parameter estimation and validation	26
1.6	Relative error	27
2	Manifolds	27
3	Cylinder	28
3.1	Cylinder flow	29
3.2	Exhaust manifold temperature	30
3.3	Engine torque	35
4	EGR-valve	36
4.1	EGR-valve mass flow	37
4.2	EGR-valve actuator	39
5	Turbocharger	41
5.1	Turbo inertia	41
5.2	Turbine	42
5.3	Compressor	47
6	Intercooler and EGR-cooler	54
7	Summary of assumptions and model equations	54
7.1	Assumptions	54
7.2	Manifolds	55
7.3	Cylinder	55
7.4	EGR-valve	56
7.5	Turbo	57
8	Model tuning and validation	59
8.1	Summary of tuning	59
8.2	Validation	61
9	Model extensions	65
9.1	Extensions: temperature states	65
9.2	Extensions: temperature states and pressure drop over inter-cooler	66
10	Conclusions	73
	References	73
A	Notation	76
2	System analysis of a Diesel Engine with VGT and EGR	79
1	Introduction	80
2	Diesel engine model	80
3	Physical intuition for system properties	81
3.1	Physical intuition for VGT position response	82
3.2	Physical intuition for EGR-valve response	83
4	Mapping of system properties	85
4.1	DC-gains	86
4.2	Zeros and a root locus	92
4.3	Non-minimum phase behaviors	95
4.4	Operation pattern for the European Transient Cycle	99
4.5	Response time	99
5	Mapping of performance variables	101

5.1	System coupling in steady state	101
5.2	Pumping losses in steady state	101
6	Conclusions	104
	References	105
A	Response time	105
B	Relative gain array	110
3	EGR-VGT Control and Tuning for Pumping Work Minimization and Emission Control	117
1	Introduction	118
2	Proposed control approach	118
2.1	Advantages of this choice	119
2.2	Control objectives	120
3	Diesel engine model	121
4	System properties	124
4.1	Steps in VGT position and EGR-valve	124
4.2	Results from an analysis of linearized diesel engine models . .	124
4.3	Pumping losses in steady state	125
5	Control structure	126
5.1	Signals, set-points, and a limit	127
5.2	Main feedback loops	128
5.3	Additional feedback loops	128
5.4	Minimizing pumping work	129
5.5	Effect of sign reversal in VGT to EGR-fraction	130
5.6	Feedforward fuel control	131
5.7	Derivative parts	132
5.8	PID parameterization and tuning	132
6	Automatic Controller Tuning	132
6.1	Solving (28)	134
7	European Transient Cycle simulations	134
7.1	Actuator oscillations	135
7.2	Balancing control objectives	136
8	Engine test cell experiments	139
8.1	Investigation of the control objectives	139
8.2	Comparison to the current production control system	142
9	Conclusions	142
	References	144
4	Controller Tuning based on Transient Selection and Optimization for a Diesel Engine with EGR and VGT	147
1	Introduction	148
1.1	Outline	148
2	Control approach	149
2.1	Control objectives	150
3	Diesel engine model	151

4	Control structure	152
4.1	Signals, set-points and a limit	152
4.2	Main feedback loops	153
4.3	Additional control modes	154
4.4	PID parameterization and implementation	154
4.5	Derivative parts	155
4.6	Fuel control	155
5	Automatic Controller Tuning	155
5.1	Cost function	156
5.2	Optimization	157
5.3	Transient selection	158
6	Results from European Transient Cycle simulations	159
6.1	Transient selection results for the European Transient Cycle .	160
6.2	Actuator oscillations	162
6.3	Balancing control objectives	162
7	Engine test cell results	166
7.1	Investigation of the control objectives	167
7.2	Results from a non-optimized transient	167
8	Conclusions	170
	References	171

5	Non-linear Compensator for handling non-linear Effects in EGR VGT Diesel Engines	175
1	Introduction	176
1.1	Control objectives	176
2	Diesel engine model	177
3	System properties	179
3.1	Mapping of sign reversal	180
4	Control structure with PID controllers	180
4.1	Engine test cell experiments	180
5	Non-linear compensator	183
5.1	Inversion of position to flow model for EGR	184
5.2	Inversion of position to flow model for EGR and VGT	186
5.3	Stability analysis of the open-loop system	187
6	Control structure with non-linear compensator	188
6.1	Main feedback loops	188
6.2	Set-point transformation and integral action	189
6.3	Saturation levels	192
6.4	Additional control modes	192
6.5	Integral action with anti-windup	194
6.6	PID parameterization and implementation	194
6.7	Stability analysis of the closed-loop system	195
7	Engine test cell experiments	195
7.1	Comparing step responses in oxygen/fuel ratio	197
7.2	Comparison on an aggressive ETC transient	197

7.3	Comparison on the complete ETC cycle	202
8	Conclusions	202
	References	203
6	Nonlinear EGR and VGT Control with Integral Action for Diesel Engines	205
1	Introduction	206
1.1	Outline	206
2	Diesel engine model	207
3	Robust nonlinear control	208
4	Control design with integral action	209
4.1	Control design model	209
4.2	Outputs and set-points	211
4.3	Integral action	211
4.4	Feedback linearization	212
4.5	Stability of the zero dynamics	213
4.6	Construction of a CLF	214
4.7	Control law	214
5	Automatic controller tuning	215
5.1	Cost function for tuning	215
5.2	Optimization	216
6	Controller evaluation	217
6.1	Benefits with integral action	219
6.2	Benefits with non-linear control and compensator	219
6.3	Importance of the non-linear compensator	221
6.4	Drawback with the proposed CLF based control design	221
6.5	Comparison on the four transient cycles	224
7	Conclusions	225
A	Analysis of stability and robustness properties for the proposed design with integral action	226
B	Analysis of stability and robustness properties for the design	227
	References	229

Part I

Introduction

Introduction

Legislated emission limits for heavy duty trucks are constantly reduced while at the same time there is a significant drive for good fuel economy. To fulfill the requirements, technologies like Exhaust Gas Recirculation (EGR) systems and Variable Geometry Turbochargers (VGT) have been introduced in diesel engines, see Fig. 1.1. The primary emission reduction mechanisms utilized are that NO_x can be reduced by increasing the intake manifold EGR-fraction and smoke can be reduced by increasing the air/fuel ratio [5]. However the EGR fraction and air/fuel ratio depend in complicated ways on the EGR and VGT actuation and it is therefore necessary to have coordinated control of the EGR and VGT to reach the legislated emission limits. Various approaches have been published, and an overview of different control aspects of diesel engines with EGR and VGT is given in [4]. A non-linear multi-variable controller based on a Lyapunov function is presented in [6], some approaches that differ in the selection of performance variables are compared in [12], and in [15] decoupling control is investigated. Other control approaches are rank one PI control [16], PI control [12], model predictive control [14], multivariable H_∞ control [11, 8], non-linear control [1], control using exhaust gas oxygen sensor [2], motion planning with model inversion [3], and feedback linearization [13].

Three structures for coordinated EGR and VGT control are here developed and investigated in an academic and industrial collaboration. The structures provide a convenient way for handling emission requirements, and the first two structures introduce a novel and straightforward approach for optimizing the engine efficiency by minimizing pumping work. Further, a non-linear compensator with PI controllers is investigated in the second structure and a non-linear control design is investigated in the third structure for handling non-linear effects. Added to that,

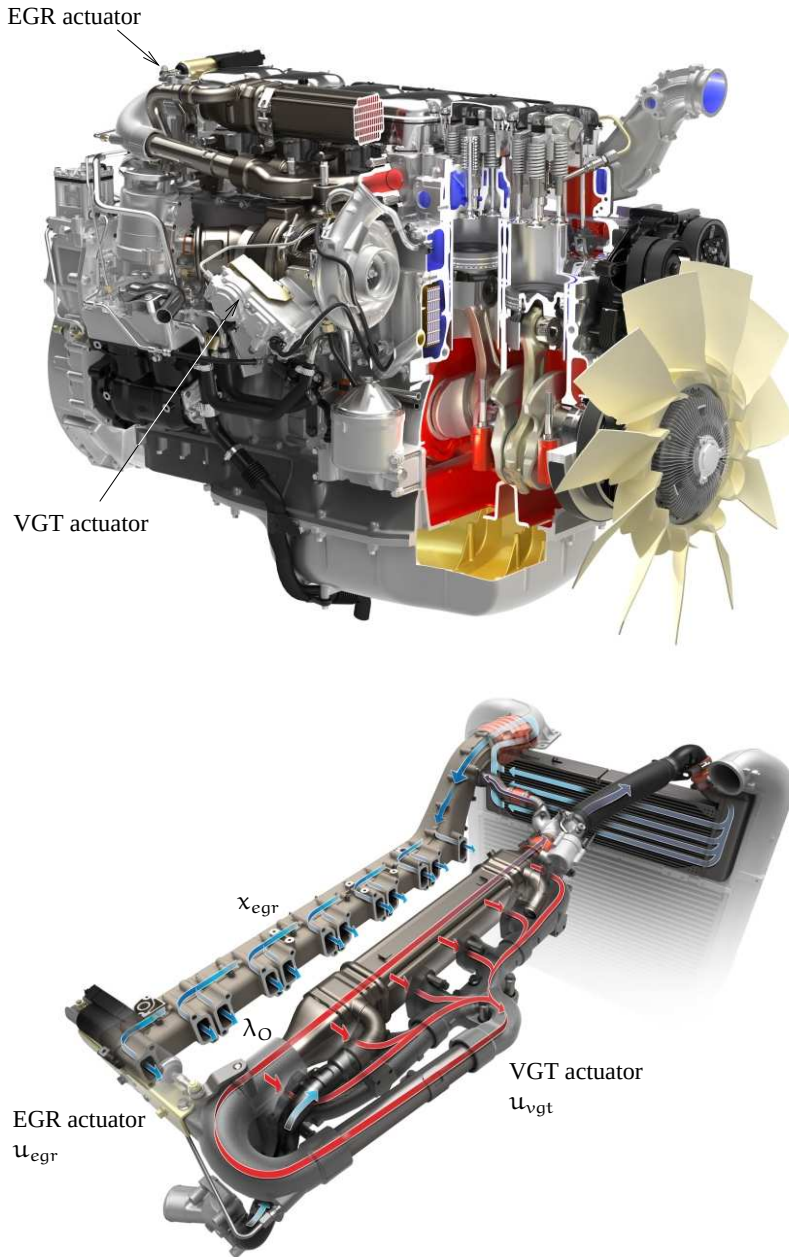


Figure 1.1 Top: Illustration of the Scania six cylinder engine with EGR and VGT used in this thesis. **Bottom:** Illustration of the EGR-system and the performance variables oxygen/fuel ratio λ_O and EGR-fraction x_{egr} used in this thesis.

the thesis covers requirements regarding additional control objectives, interfaces between inner and outer loops, and calibration that have been important for industrial validation and application.

The selection of performance variables is an important first step [19], and for emission control it should be noted that exhaust gases, present in the intake from EGR, also contain oxygen. This makes it more suitable to define and use the oxygen/fuel ratio instead of the traditional air/fuel ratio. The main motive is that it is the oxygen content that is crucial for smoke generation, and the idea is to use the oxygen content of the cylinder instead of air mass flow, see e.g. [10]. Thus, intake manifold EGR-fraction x_{egr} and oxygen/fuel ratio λ_{O} in the cylinder (see Fig. 1.1) are a natural selection for performance variables as they are directly related to the emissions. These performance variables are equivalent to cylinder air/fuel ratio and burned gas ratio which are a frequent choice for performance variables [6, 12, 13, 16].

The main goal of this thesis is to design control structures that regulate the performance variables x_{egr} and λ_{O} by using the EGR and VGT actuators.

The publications related to this thesis will be described in Sec. 1.1. Sec. 1.2 will give an overview and describe the contributions of the six publications presented in this thesis.

1.1 List of Publications

This thesis is based on the following publications

- Publication 1 is also available as the technical report "Modeling of a Diesel Engine with VGT and EGR capturing sign reversal and non-minimum phase behaviors" by Johan Wahlström and Lars Eriksson. An earlier version of this material was presented in the technical report "Modeling of a diesel engine with VGT and EGR including oxygen mass fraction" by Johan Wahlström and Lars Eriksson, and in the Licentiate thesis "Control of EGR and VGT for emission control and pumping work minimization in diesel engines" by Johan Wahlström.
- Publication 2 is also available as the technical report "System analysis of a Diesel Engine with VGT and EGR" by Johan Wahlström, Lars Eriksson, and Lars Nielsen. An earlier version of this material was presented in the Licentiate thesis "Control of EGR and VGT for emission control and pumping work minimization in diesel engines" by Johan Wahlström.
- Publication 3 has been submitted for publication. Parts of this material were presented in the Licentiate thesis "Control of EGR and VGT for emission control and pumping work minimization in diesel engines" by Johan Wahlström. Related to this publication is the conference paper "PID controllers and their tuning for EGR and VGT control in diesel engines" by Johan Wahlström, Lars Eriksson, Lars Nielsen, and Magnus Pettersson, 16th

IFAC World Congress, 2005, that proposes a control structure that is similar to the control structure in Publication 3.

- Publication 4 has been published as the conference paper "Controller tuning based on transient selection and optimization for a diesel engine with EGR and VGT" by Johan Wahlström, Lars Eriksson, and Lars Nielsen, SAE Technical paper 2008-01-0985, Detroit, USA, 2008. Parts of this material were presented in the Licentiate thesis "Control of EGR and VGT for emission control and pumping work minimization in diesel engines" by Johan Wahlström".
- Publication 5 is also available as the technical report "Non-linear compensator for handling non-linear Effects in EGR VGT Diesel Engines" by Johan Wahlström and Lars Eriksson. Related to this publication is the conference paper "Performance gains with EGR-flow inversion for handling non-linear dynamic effects in EGR VGT CI engines" by Johan Wahlström and Lars Eriksson, Fifth IFAC Symposium on Advances in Automotive Control, 2007.
- An earlier version of Publication 6 has been published as the conference paper "Robust Nonlinear EGR and VGT Control with Integral Action for Diesel Engines" by Johan Wahlström and Lars Eriksson, 17th IFAC World Congress, 2008.

1.2 Overview and Contributions of the Publications

An overview of the six publications in this thesis is presented below and for each publication its contributions.

1.2.1 Publication 1 - Modeling

When developing and validating a controller for a diesel engine with VGT and EGR, it is desirable to have a model that describes the system dynamics and the nonlinear effects. Therefore, the objective of Publication 1 is to construct a mean value diesel engine model with VGT and EGR. For these systems, several models with different selections of states and complexity have been published [1, 6, 7, 9, 16, 18].

Here the model should be able to describe stationary operations and dynamics that are important for gas flow control. The intended applications of the model are system analysis, simulation, and development of model-based control systems. The goal is to construct a model that describes the dynamics in the manifold pressures, turbocharger, EGR, and actuators with few states in order to have short simulation times. Therefore the model has only eight states: intake and exhaust manifold pressures, oxygen mass fraction in the intake and exhaust manifold, turbocharger speed, and three states describing the actuator dynamics. The structure of the model can be seen in Fig. 1.2. The model is more complex than e.g. the third

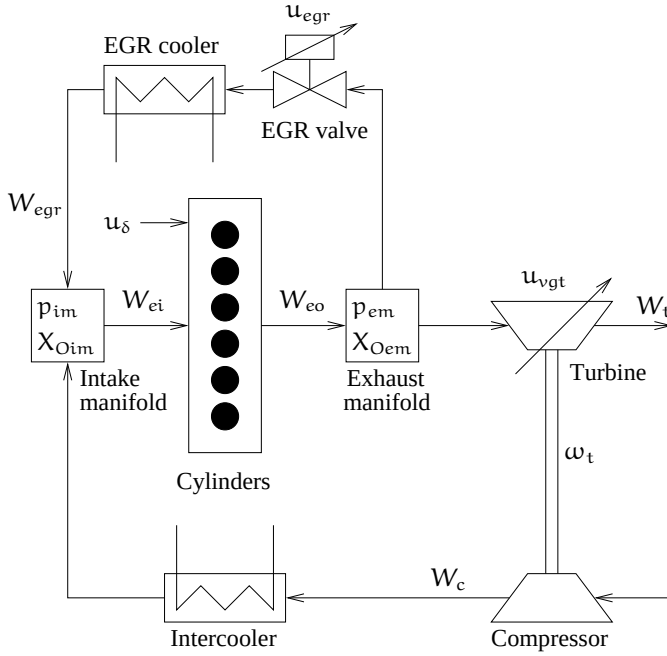


Figure 1.2 A model structure of the diesel engine. It has three control inputs and five main states related to the engine (p_{im} , p_{em} , X_{Oim} , X_{Oem} , and ω_t). In addition, there are three states for actuator dynamics ($\tilde{u}_{egr1}$, $\tilde{u}_{egr2}$, and $\tilde{u}_{vgt}$).

order model in [6] that only describes the pressure and turbocharger dynamics, but it is considerably less complex than a GT-POWER model that is based on one-dimensional gas dynamics [17].

Many models in the literature, that have approximately the same complexity as the model proposed here, use three states for each control volume in order to describe the temperature dynamics [6, 9, 16]. However, the model proposed here uses only two states for each manifold. Model extensions are investigated showing that inclusion of temperature states and pressure drop over the intercooler only have minor effects on the dynamic behavior in pressure, oxygen mass fraction, and turbocharger speed and does not improve the model quality. Therefore, these extensions are not included in the proposed model.

Model equations and tuning methods are described for each subsystem in the model. In order to have a low number of model parameters, flows and efficiencies are modeled using physical relationships and parametric models instead of look-up tables. To tune and validate the model, stationary and dynamic measurements have been performed in an engine laboratory at Scania CV AB. Static and dynamic validations of the entire model using dynamic experimental data show that the

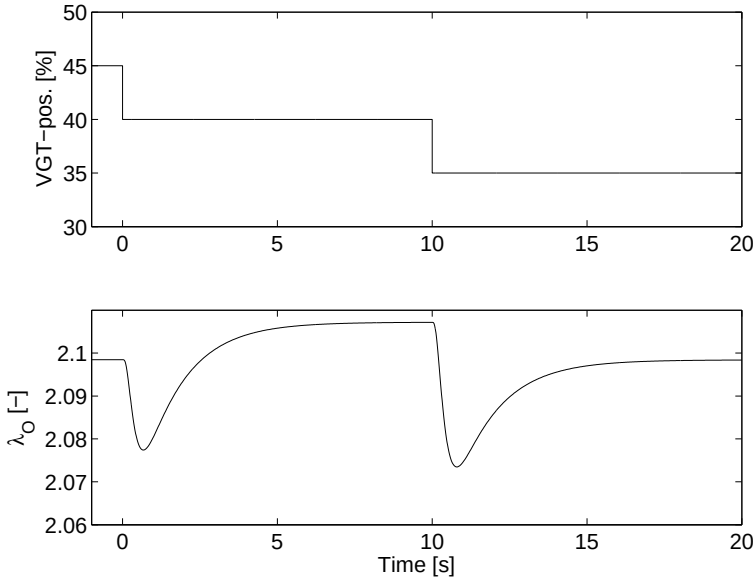


Figure 1.3 Non-minimumphase behavior and sign reversal in the channel VGT-position to λ_O . The DC-gain in the first step is negative and the DC-gain in the second step is positive.

mean relative errors are 12.7 % or lower for all measured variables. The validations also show that the proposed model captures the essential system properties, i.e. a non-minimum phase behavior in the channel $u_{egr} \rightarrow p_{im}$ and a non-minimum phase behavior, an overshoot, and a sign reversal in the channel $u_{vgt} \rightarrow W_c$.

1.2.2 Publication 2 - System analysis

An analysis of the characteristics and the behavior of a system aims at obtaining insight into the control problem. This is known to be important for a successful design of an EGR and VGT controller due to non-trivial intrinsic properties, see for example [9]. Therefore, the goal is to make a system analysis of the diesel engine model proposed in Publication 1.

Step responses over the entire operating region show that the channels $u_{vgt} \rightarrow \lambda_O$, $u_{egr} \rightarrow \lambda_O$, and $u_{vgt} \rightarrow x_{egr}$ have non-minimum phase behaviors and sign reversals. See for example Fig. 1.3 that shows these system properties for $u_{vgt} \rightarrow \lambda_O$. The fundamental physical explanation of these system properties is that the system consists of two dynamic effects that interact: a fast pressure dynamics in the manifolds and a slower turbocharger dynamics. It is shown that the engine frequently operates in operating points where the non-minimum phase behaviors and sign reversals occur for the channels $u_{vgt} \rightarrow \lambda_O$ and $u_{vgt} \rightarrow x_{egr}$, and consequently, it is

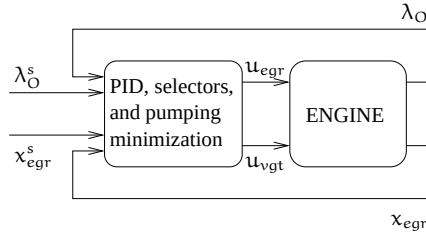


Figure 1.4 A control structure with PID controllers, min/max selectors, and pumping minimization. It handles the sign reversal in Fig. 1.3 by avoiding the loop VGT-position to λ_O .

important to consider these properties in a control design. Further, an analysis of zeros for linearized multiple input multiple output models of the engine shows that they are non-minimum phase over the complete operating region. A mapping of the performance variables λ_O and x_{egr} and the relative gain array show that the system from u_{egr} and u_{vgt} to λ_O and x_{egr} is strongly coupled in a large operating region. It is also illustrated that the pumping losses $p_{em} - p_{im}$ decrease with increasing EGR-valve and VGT opening for almost the complete operating region.

1.2.3 Publication 3 - EGR-VGT Control for Pumping Work Minimization

A control structure with PID controllers and selectors (see Fig. 1.4) is proposed and investigated for coordinated control of oxygen/fuel ratio λ_O and intake manifold EGR-fraction x_{egr} . These were chosen both as performance and feedback variables since they give information about when it is possible to minimize the pumping work. This pumping work minimization is a novel and simple strategy and compared to another control structure which closes the EGR-valve and the VGT more, the pumping work is substantially reduced. Further, the chosen variables are strongly coupled to the emissions which makes it easy to adjust set-points, e.g. depending on measured emissions during an emission calibration process. This is more straightforward than control of manifold pressure and air mass flow which is a common choice of feedback variables in the literature [8, 11, 12, 15, 16]. Other choices of feedback variables in the literature are intake manifold pressure and EGR-fraction [12], exhaust manifold pressure and compressor air mass flow [6], intake manifold pressure and EGR flow [14], intake manifold pressure and cylinder air mass-flow [1], or compressor air mass flow and EGR flow [3].

Based on the system analysis in Publication 2, λ_O is controlled by the EGR-valve and x_{egr} by the VGT-position, mainly to handle the sign reversal from VGT to λ_O in Fig. 1.3.

Besides controlling the two main performance variables, λ_O and x_{egr} , the control structure also successfully handles torque control, including torque limitation

due to smoke control, and supervisory control of turbo charger speed for avoiding over-speeding. Further, the control objectives are mapped to the controller structure via a systematic analysis of the control problem, and this conceptual coupling to objectives gives the foundation for systematic tuning. This is utilized to develop an automatic controller tuning method. The objectives to minimize pumping work and ensure the minimum limit of λ_O are handled by the structure, while the other control objectives are captured in a cost function, and the tuning is formulated as a non-linear least squares problem. The details of the tuning method are described in Publication 4.

Different performance trade-offs are necessary and they are illustrated on the European Transient Cycle. The proposed controller is validated in an engine test cell, where it is experimentally demonstrated that the controller achieves all control objectives and that the current production controller has at least 26% higher pumping losses compared to the proposed controller.

1.2.4 Publication 4 - Controller Tuning

Efficient calibration is important and as mentioned above a control tuning method has been developed. The proposed tuning method is based on control objectives that are captured in a cost function, and the tuning is formulated as a non-linear least squares problem. The method is illustrated by applying it on the control structure in Publication 3 and it is also used for the control structures in Publication 5 and 6. To aid the tuning, a systematic method is developed for selecting significant transients that exhibit different challenges for the controller, and an important step in obtaining the solution is precautions in a separate phase to avoid ending up in an unsatisfactory local minimum.

The performance is evaluated on the European Transient Cycle. It is demonstrated how the weights in the cost function influence behavior, and that the tuning method is important in order to improve the control performance compared to if only the initialization method is used. Furthermore, it is shown that the control structure in Publication 3 with parameters based on the proposed tuning method achieves all the control objectives, and it is successfully applied in an engine test cell.

The most important sections in Publication 4 is the automatic tuning method in Sec. 5 and the simulation results in Sec. 6. The control approach in Sec. 2, the control structure in Sec. 4, and the experimental validations in Sec. 7 are more completely described in Publication 3. The simulations in Publication 3 and 4 are performed with an earlier version of the model in Publication 1 that only has two states for the actuator dynamics. However, simulations with the model in Publication 1 that has three states for the actuator dynamics have been performed showing the same results as the results in Publication 3 and 4.

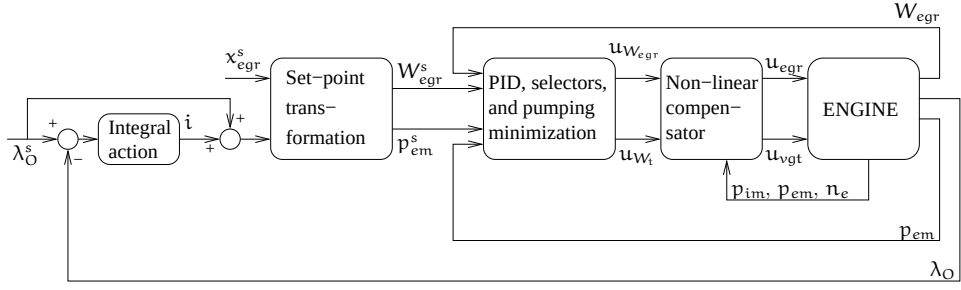


Figure 1.5 The control structure in Fig. 1.4 is extended with a non-linear compensator.

1.2.5 Publication 5 - Non-linear compensator

Inspired by an approach in [6], the control structure in Fig. 1.4 is extended with a non-linear compensator according to Fig. 1.5. The goal is to investigate if this non-linear compensator improves the control performance compared to the controller in Fig. 1.4. The non-linear compensator is a non-linear state dependent input transformation that is developed by inverting the models, for EGR-flow and turbine flow, that have actuator position as input and flow as output. This leads to two new control inputs, $u_{W_{egr}}$ and u_{W_t} , which are equal to the EGR-flow W_{egr} and the turbine flow W_t if there are no model errors in the non-linear compensator.

A system analysis of the open-loop system with a non-linear compensator shows that it handles sign reversals and non-linear effects. Further, the analysis shows that this open-loop system is unstable in a large operating region. This instability is stabilized by a control structure that consists of PID controllers, min/max-selectors, and a pumping minimization mechanism similar to the structure in Fig. 1.4. The EGR flow W_{egr} and the exhaust manifold pressure p_{em} are chosen as feedback variables in this structure. Further, the set-points for EGR-fraction and oxygen/fuel ratio are transformed to set-points for the feedback variables. In order to handle model errors in this set-point transformation, an integral action on λ_O is used in an outer loop. Experimental validations of the control structure in Fig. 1.5 show that it handles nonlinear effects (see Fig. 1.6), and that it reduces EGR-errors but increases the pumping losses compared to the control structure in Fig. 1.4.

1.2.6 Publication 6 - Non-linear control

A non-linear controller based on a design in [6] that utilizes a control Lyapunov function and inverse optimal control is investigated. The feedback variables are compressor flow W_c and exhaust manifold pressure p_{em} , see Fig. 1.7. The PID controllers in Fig. 1.5 are thus replaced by a non-linear multivariable controller according to Fig. 1.7, and the goal is to investigate if this non-linear controller improves the control performance compared to the controller in Fig. 1.5. Simulations

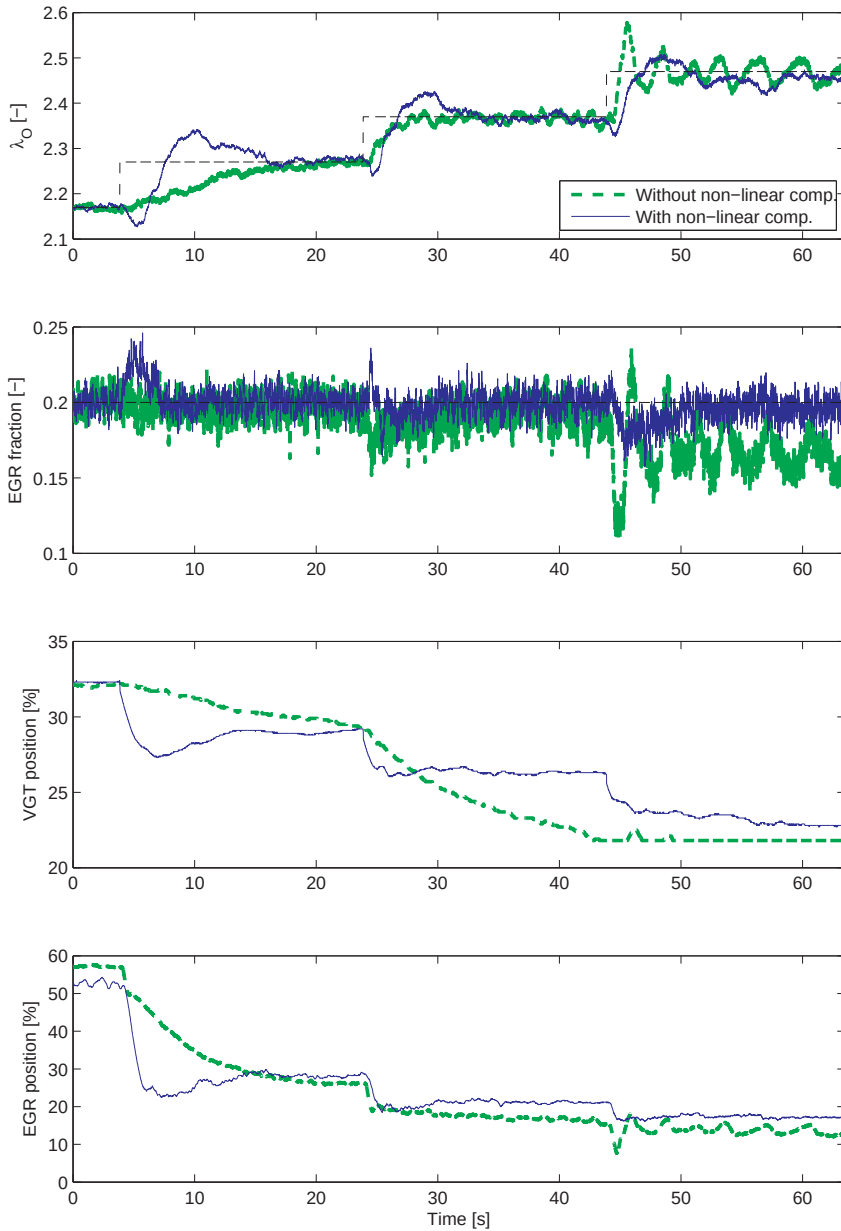


Figure 1.6 The control structure without non-linear compensator (Fig. 1.4) gives slow control and oscillations at different steps, i.e. it does not handle non-linear effects. The control structure with non-linear compensator (Fig. 1.5) gives less oscillations and fast control, i.e. it handles nonlinear effects.

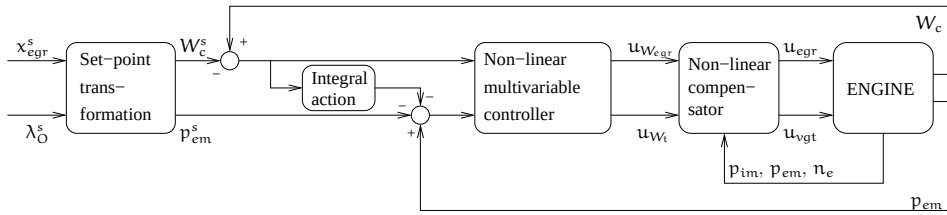


Figure 1.7 The PID controllers in Fig. 1.5 are replaced by a non-linear multivariable controller that is based on a Lyapunov function and inverse optimal control. Simulations show that this design is not robust to model errors in the non-linear compensator while the control structure in Fig. 1.5 is. If there are no model errors in the non-linear compensator Fig. 1.5 and 1.7 have approximately the same control performance.

show that integral action is necessary to handle model errors, so the design in [6] is extended with integral action on the compressor flow W_c as depicted in Fig. 1.7 so that the controller can track the performance variables specified in the outer loop.

Comparisons by simulation show that the proposed control design handles non-linear effects in the diesel engine, and that the non-linear compensator is important to achieve this. If there are no model errors in the non-linear compensator, the controllers in Fig. 1.5 and 1.7 have approximately the same control performance. However, it is shown that the proposed control design in Fig. 1.7 is not robust to model errors in the non-linear compensator while the control structure in Fig. 1.5 is, and due to these results, the control structure in Publication 6 is not experimentally validated. Instead, the control structure in Publication 5 is recommended.

Bibliography

- [1] M. Ammann, N.P. Fekete, L. Guzzella, and A.H. Glattfelder. Model-based Control of the VGT and EGR in a Turbocharged Common-Rail Diesel Engine: Theory and Passenger Car Implementation. *SAE Technical paper 2003-01-0357*, January 2003.
- [2] A. Amstutz and L. Del Re. EGO sensor based robust output control of EGR in diesel engines. *IEEE Transactions on Control System Technology*, pages 37–48, 1995.
- [3] Jonathan Chauvin, Gilles Corde, Nicolas Petit, and Pierre Rouchon. Motion planning for experimental airpath control of a diesel homogeneous charge-compression ignition engine. *Control Engineering Practice*, 2008.
- [4] L. Guzzella and A. Amstutz. Control of diesel engines. *IEEE Control Systems Magazine*, 18:53–71, 1998.

- [5] J.B. Heywood. *Internal Combustion Engine Fundamentals*. McGraw-Hill Book Co, 1988.
- [6] M. Jankovic, M. Jankovic, and I.V. Kolmanovsky. Constructive lyapunov control design for turbocharged diesel engines. *IEEE Transactions on Control Systems Technology*, 2000.
- [7] M. Jung. *Mean-Value Modelling and Robust Control of the Airpath of a Turbocharged Diesel Engine*. PhD thesis, University of Cambridge, 2003.
- [8] Merten Jung, Keith Glover, and Urs Christen. Comparison of uncertainty parameterisations for H-infinity robust control of turbocharged diesel engines. *Control Engineering Practice*, 2005.
- [9] I.V. Kolmanovsky, A.G. Stefanopoulou, P.E. Moraal, and M. van Nieuwstadt. Issues in modeling and control of intake flow in variable geometry turbocharged engines. In *Proceedings of 18th IFIP Conference on System Modeling and Optimization*, Detroit, July 1997.
- [10] Shigeki Nakayama, Takao Fukuma, Akio Matsunaga, Teruhiko Miyake, and Toru Wakimoto. A new dynamic combustion control method based on charge oxygen concentration for diesel engines. In *SAE Technical Paper 2003-01-3181*, 2003. SAE World Congress 2003.
- [11] M. Nieuwstadt, P.E. Moraal, I.V. Kolmanovsky, A. Stefanopoulou, P. Wood, and M. Widdle. Decentralized and multivariable designs for EGR–VGT control of a diesel engine. In *IFAC Workshop, Advances in Automotive Control*, 1998.
- [12] M.J. Nieuwstadt, I.V. Kolmanovsky, P.E. Moraal, A.G. Stefanopoulou, and M. Jankovic. EGR–VGT control schemes: Experimental comparison for a high-speed diesel engine. *IEEE Control Systems Magazine*, 2000.
- [13] R. Rajamani. Control of a variable-geometry turbocharged and wastegated diesel engine. *Proceedings of the I MECH E Part D Journal of Automobile Engineering*, November 2005.
- [14] J. Rückert, F. Richert, A. Schloßer, D. Abel, O. Herrmann, S. Pischinger, and A. Pfeifer. A model based predictive attempt to control boost pressure and EGR–rate in a heavy duty diesel engine. In *IFAC Symposium on Advances in Automotive Control*, 2004.
- [15] J. Rückert, A. Schloßer, H. Rake, B. Kinoo, M. Krüger, and S. Pischinger. Model based boost pressure and exhaust gas recirculation rate control for a diesel engine with variable turbine geometry. In *IFAC Workshop: Advances in Automotive Control*, 2001.
- [16] A.G. Stefanopoulou, I.V. Kolmanovsky, and J.S. Freudenberg. Control of variable geometry turbocharged diesel engines for reduced emissions. *IEEE Transactions on Control Systems Technology*, 8(4), July 2000.

-
- [17] Gamma Technologies. *GT-POWER User's Manual 6.1*. Gamma Technologies Inc, 2004.
 - [18] C. Vigild. *The Internal Combustion Engine Modelling, Estimation and Control Issues*. PhD thesis, Technical University of Denmark, Lyngby, 2001.
 - [19] Kemin Zhou, John C. Doyle, and Keith Glover. *Robust and optimal control*. Prentice-Hall, Inc., Upper Saddle River, NJ, USA, 1996.